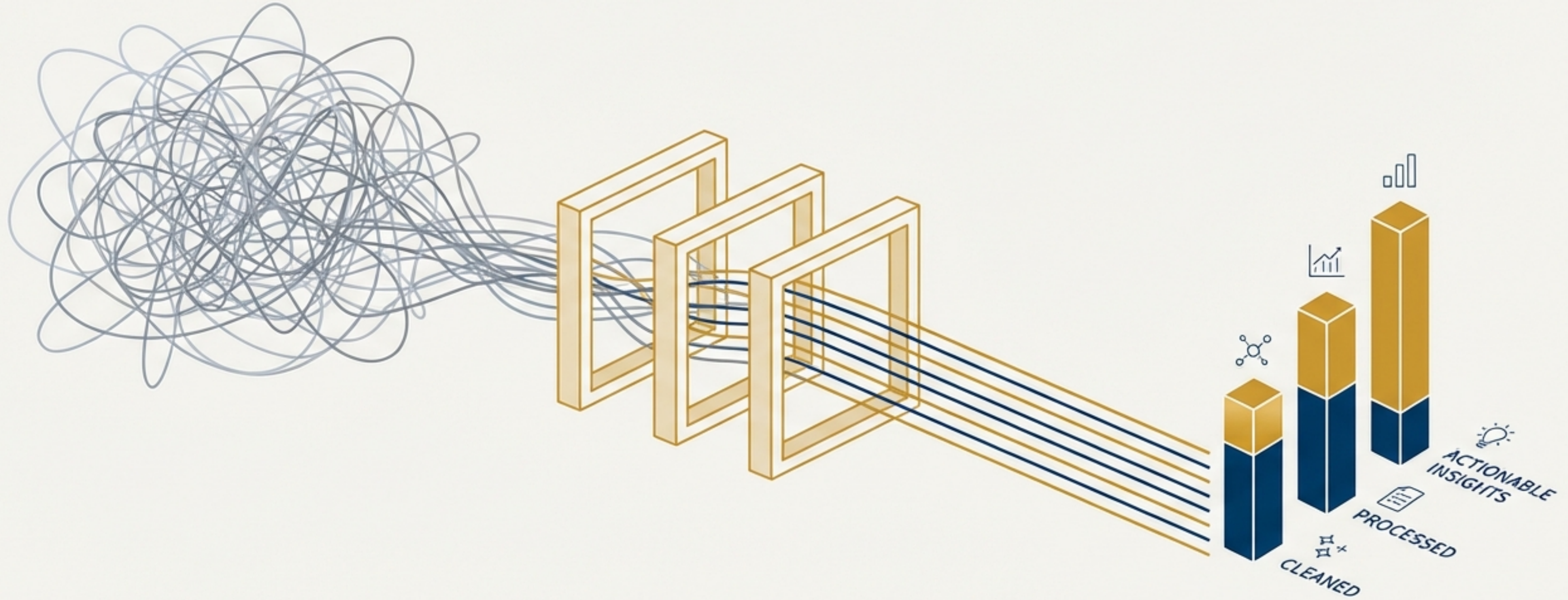
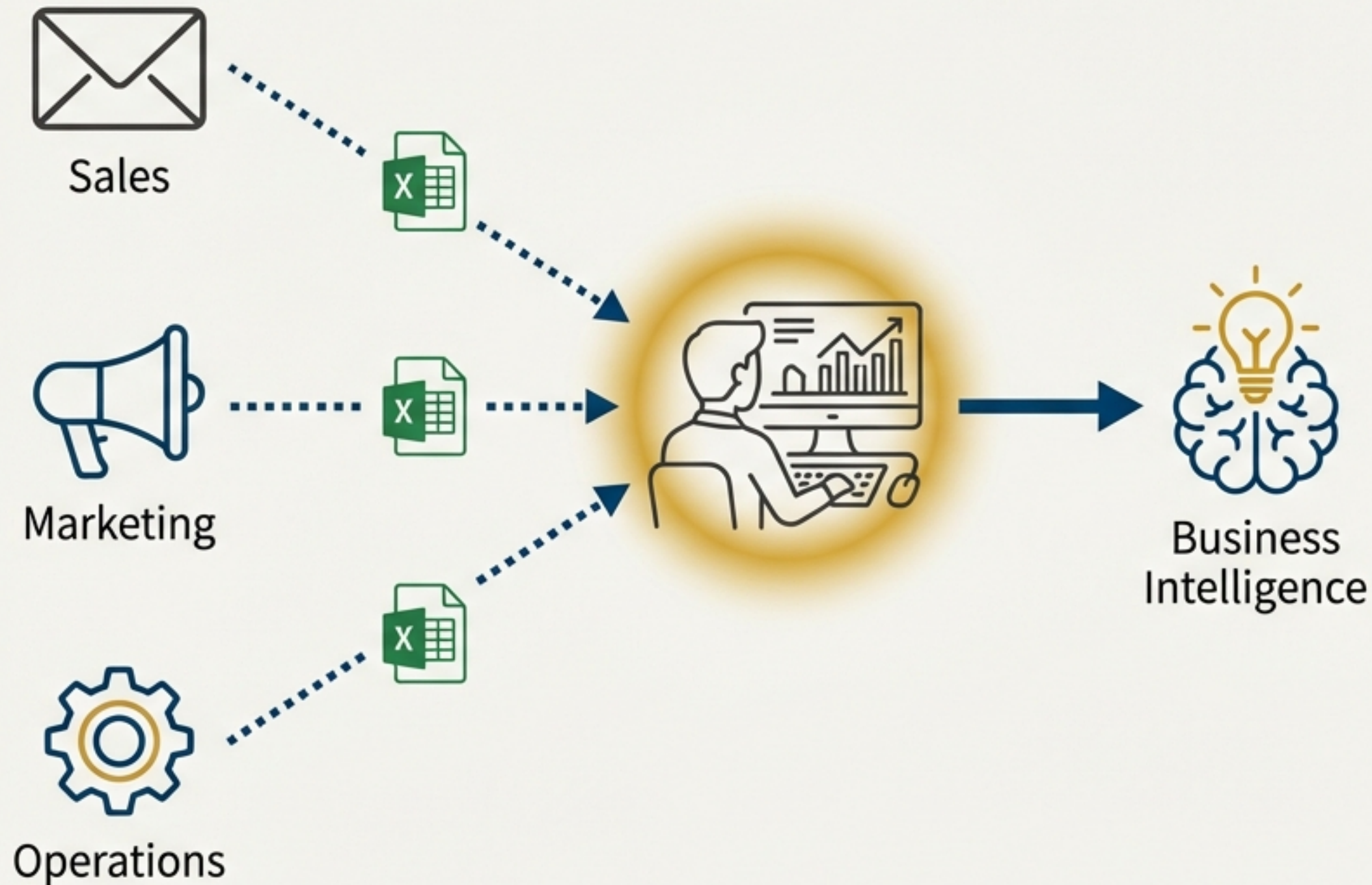


From Raw Data to Strategic Insight

A Blueprint for an End-to-End Data Analysis Workflow



The Daily Challenge: Turning Disparate Data into Business Intelligence



Source:

Daily and weekly Excel files arrive via email from Sales, Marketing, and Operations teams.

Role:

You are the Data Analyst responsible for the entire data lifecycle.

Objective:

Validate, clean, integrate, and analyse this data to provide actionable insights to senior leadership (CEO, Directors).

Establishing Order: The Foundation of a Trustworthy Data Pipeline

Systematic Data Receipt

- **Action:** Daily monitoring of the sales mailbox to identify and retrieve new data files.
- **Source Teams:** Sales, Marketing, Operations.

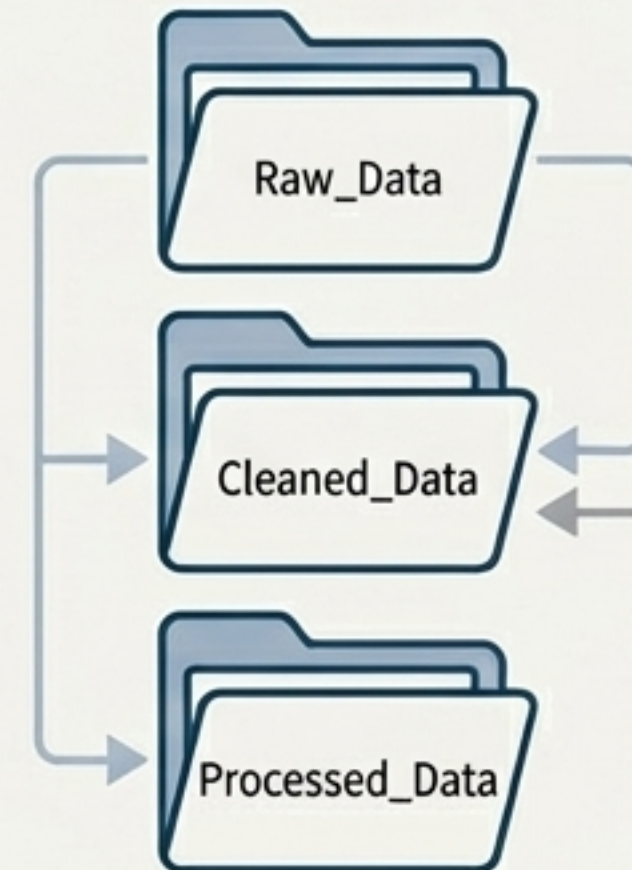
Structured File Management

- **Action:** Download all Excel files and save them into a predefined, logical folder system. This prevents data loss and ensures process integrity.

```
Data_Analysis/  
├── Raw_Data/  
│   ├── Sales_2024_01.xlsx  
│   └── Sales_2024_02.xlsx  
├── Cleaned_Data/  
└── Processed_Data/
```

Why This Matters

A disciplined file management system is the bedrock of any reliable data process. It ensures traceability, prevents errors, and makes automation possible.



The First Line of Defence: Initial Validation and Cleaning in Excel

File Validation Checks (Before Loading)

- ✓ **File Format:** Is it the correct `.xlsx` format?
- ✓ **Column Integrity:** Are all expected columns present?
- ✓ **Data Types:** Do columns contain the correct data types (date, number, text)?
- ✓ **File Health:** Is the file free from corruption or errors?

Data Cleaning Activities



Remove extraneous rows (e.g., repeated headers, totals).



Correct spelling inconsistencies (e.g., Country names, Product names).



Standardise formats for consistency (e.g., uniform date format, proper text case).

Country	Date	Sales
london	01/03/24	1000
---	---	---
London	Mar-02-2024	1500
	03-03-24	2000
Total		4500

Before



Country	Date	Sales
London	2024-03-01	1000
London	2024-03-02	1500
London	2024-03-03	2000

After

Profiling and Mapping: Understanding the Data's DNA

Data Profiling – What Does the Data Tell Us?



Count number of rows and columns.



Identify unique values per column.



Determine Min/Max values for numerical fields.



Analyse the overall distribution of data.

Key questions answered by profiling:

Is the data usable? Is anything missing?
Are there abnormal values?

Data Mapping – Aligning Source to Target

Purpose: Ensures consistency before loading into the database by aligning source column names with the standardised target schema.



Source Column	Target Column
City	City_Name
Or_Date	Order_Date
Rest_Name	Restaurant_Name



The Quality Gate: A Zero-Tolerance Approach to Data Issues

All identified issues must be fixed *before* loading data into the database.



Missing Values (Nulls)

- ✓ Identify empty cells in mandatory fields (State, City, Order_Date).
- ✓ Decide whether to fill with a default, remove the record, or permit NULL.



Blank/Empty Strings

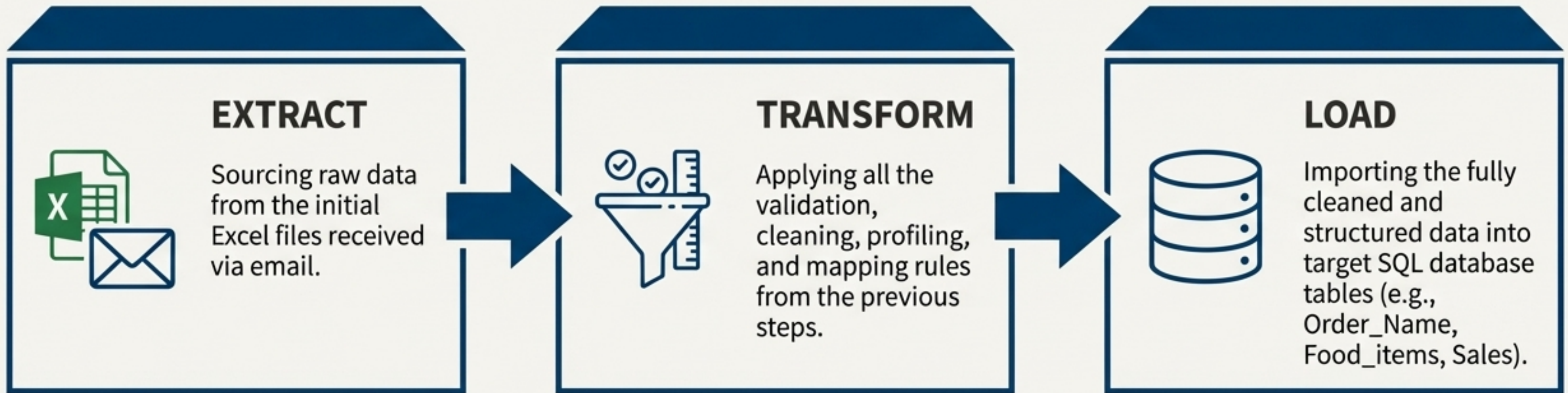
Detect records where crucial fields contain empty text rather than NULL values.



Duplicate Records

Implement checks to detect and remove duplicate entries to ensure data integrity.

Centralising a Single Source of Truth: The ETL Process



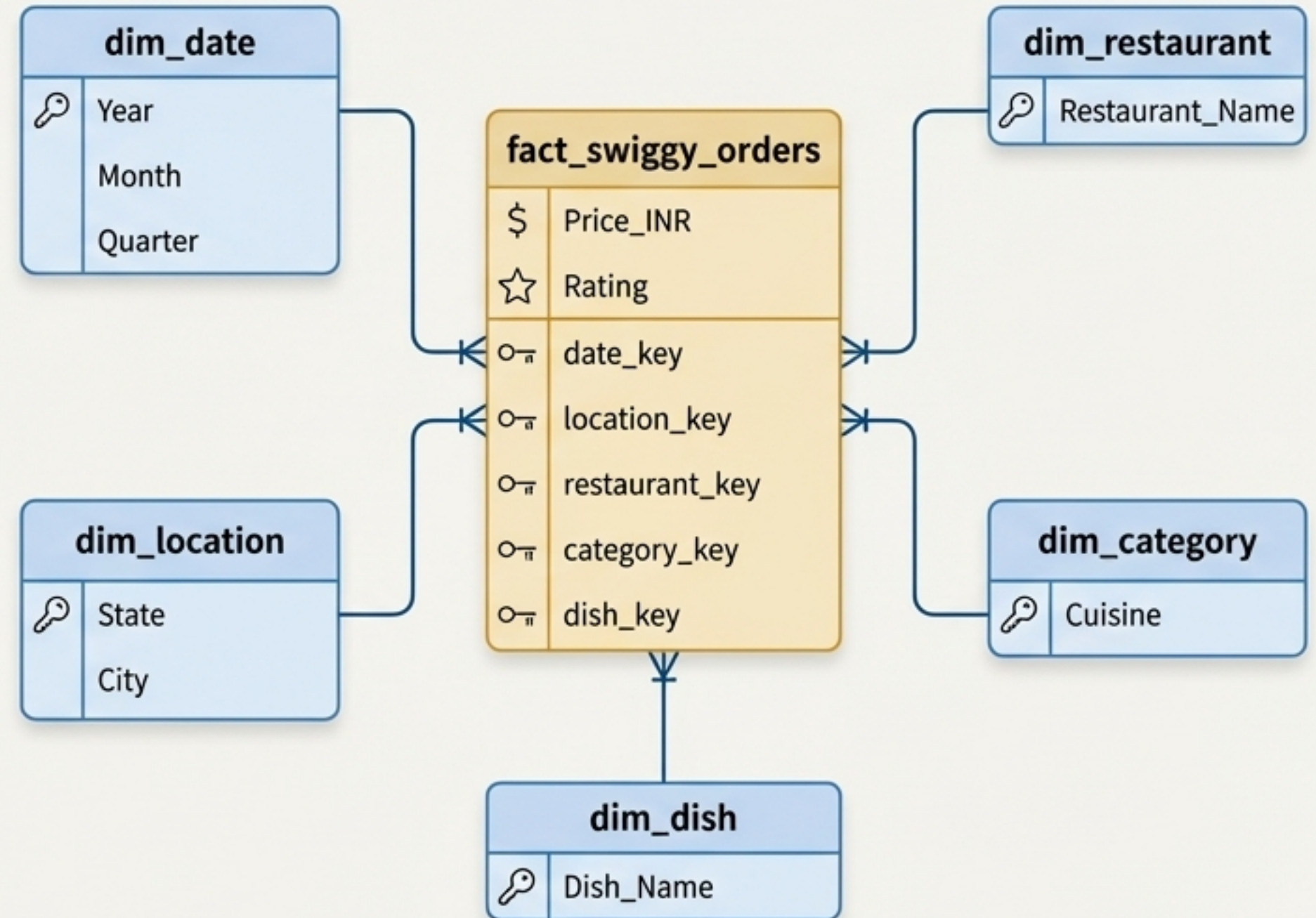
Strategic Outcome: The ETL process creates a centralized and **trusted** data repository. This is the **single source of truth** for all subsequent analysis and **business** intelligence.

Designing for Performance and Clarity: The Star Schema

Why Dimensional Modelling?

Dimensional modelling organises data to make analysis simple, consistent, and efficient. It separates descriptive information (Dimensions) from measurable values (Facts), reducing data duplication and improving query performance.

Most BI tools are optimised for this structure, ensuring a scalable and reliable foundation for analytics.



Ensuring Database Integrity and Data Privacy

Post-Load SQL Validation

Even after loading, a final layer of checks is performed in SQL to guarantee integrity.

- ✓ **Row Count Verification:** Does the table row count match the source?
- ✓ **Foreign Key Validation:** Are all relationships intact?
- ✓ **Null Checks:** Re-verify critical columns (``SELECT COUNT(*) FROM sales WHERE customer_id IS NULL;``).

Responsible Data Masking

Management needs insights, not sensitive personal data. Data masking is essential for privacy and compliance.

Mask sensitive fields like Email ID, Customer Name, and Phone Number.

Before: "john.doe@gmail.com"
After Masking: "jXXXX@XXXX.com"



From Data to Decisions: Unleashing SQL for Analysis

Developing Key Performance Indicators (KPIs)

Total Orders

1.2M



Total Revenue

₹450M



Average Dish Price

₹285



Average Rating

4.1 ★



Powering Deep-Dive Business Analysis

Move beyond high-level metrics to answer complex business questions.

Example SQL Query

```
-- Calculate revenue by country
SELECT
    country,
    SUM(quantity * price) AS revenue
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
GROUP BY country;
```

A Multi-Dimensional View of Business Performance



Date-Based Analysis

- Monthly & quarterly order trends
- Year-wise growth patterns
- Day-of-week performance analysis



Location-Based Analysis

- Top 10 cities by order volume
- Revenue contribution by state



Food & Restaurant Performance

- Top 10 restaurants by orders
- Most ordered dishes and categories
- Cuisine performance (Orders vs. Average Rating)



Customer Spending Insights

- Distribution of orders across spending buckets:
 - Under 100, 100–199, 200–299, 300–499, 500+

Bridging the Gap: Translating SQL Analysis for Business Users

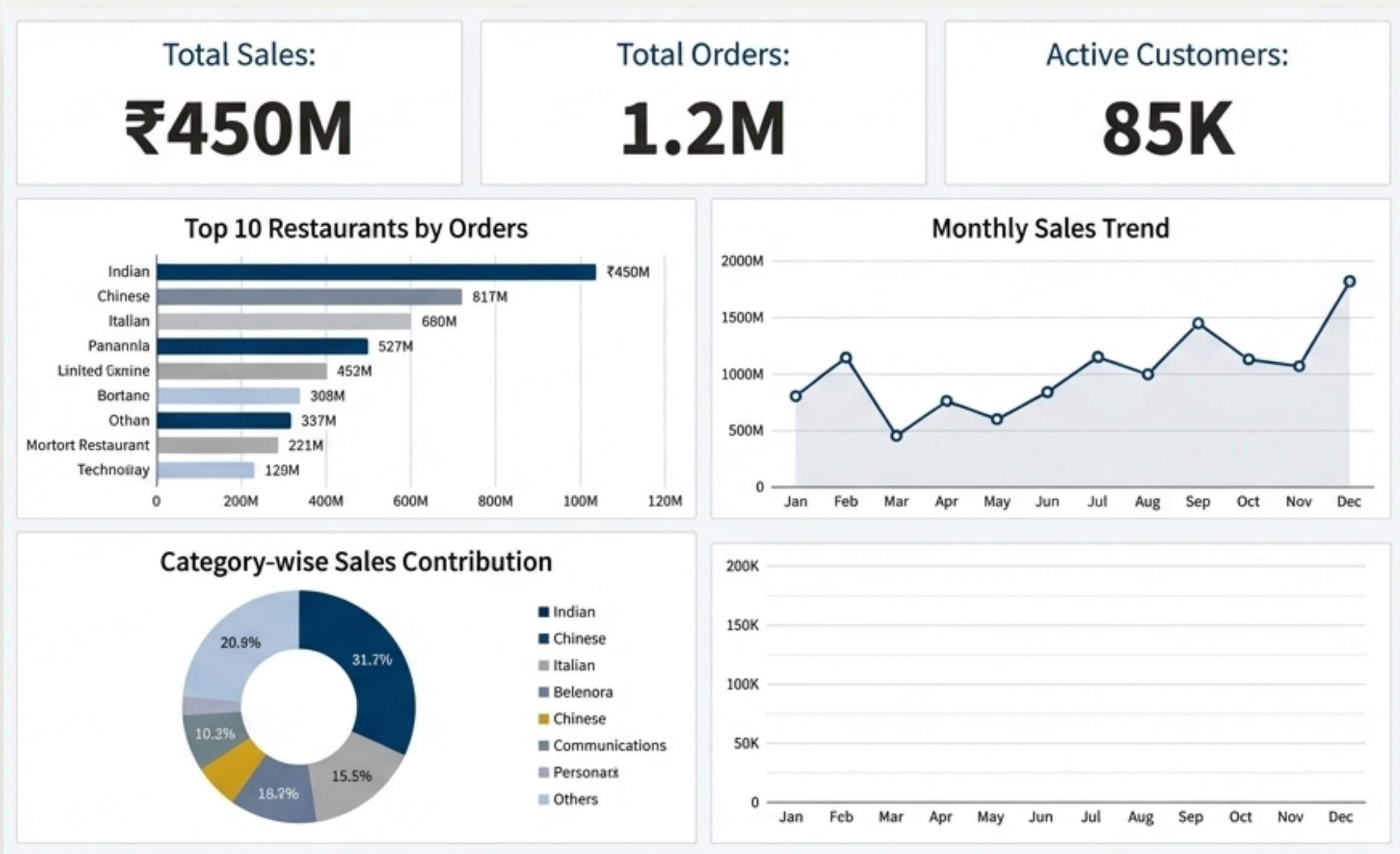


Why Excel?

- Empowers stakeholders with a familiar tool.
- Enables rapid ad-hoc analysis via **Pivot Tables**.
- Facilitates the creation of **Summary Sheets** and **Charts** for reporting.

This step transforms technical output into a flexible asset for the entire business.

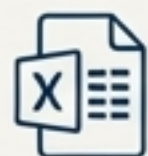
Visualising the Narrative: The Excel Dashboard



The final dashboard provides at-a-glance insights, allowing stakeholders to quickly understand performance and identify trends.

Delivering Actionable Insights to Leadership

The Final Package:



The Excel Dashboard: The primary visual tool for exploration.



Summary Insights: A concise list of key findings, written in clear business language (bullet points).



Strategic Recommendations: Actionable suggestions based on the data-driven insights.

The Audience:



CEO



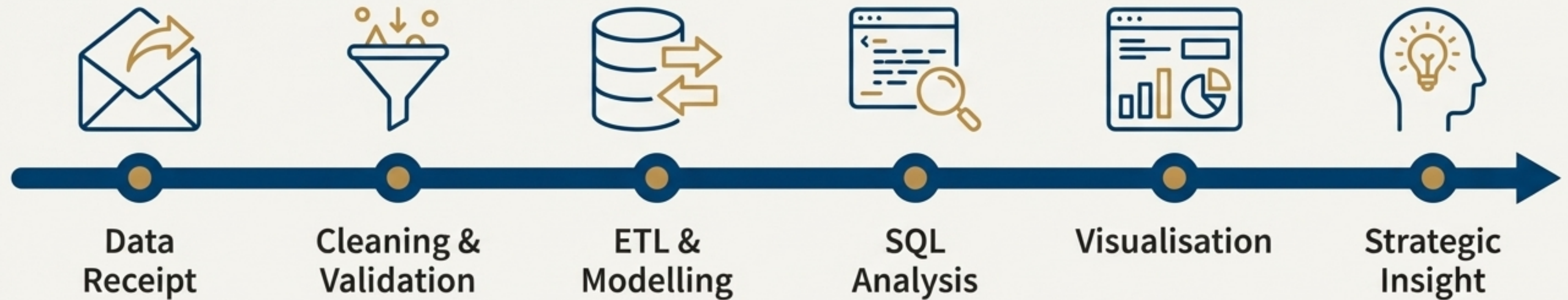
Director



Sales Head



The Analyst's Contribution: The Complete Data Value Chain



“ I handle end-to-end data management — from receiving raw Excel files via email, validating and cleaning data, performing ETL, integrating data into SQL databases, running analysis queries, creating Excel dashboards, and finally sharing actionable insights with stakeholders. ”